Data Analyst Capstone Project Report

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OUTLINE



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- Introduction
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- Discussion
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EXECUTIVE SUMMARY



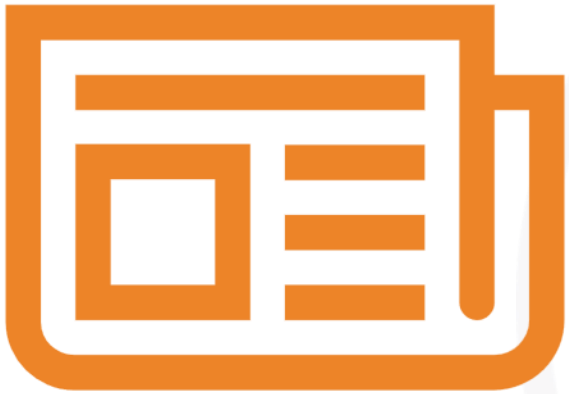
- Objective: Analyse developer technology trends, job opportunities and workforce demographics.
- Data Sources: IBM Jobs API, programming-language salary data and Stack Overflow Developer Survey.
- Methods: Data cleaning, exploratory analysis, Python visualisations and three Looker Studio dashboards.
- Key Findings: JavaScript leads current language usage; Python leads desired languages; PostgreSQL leads current and desired database selections.
- Compensation: Median reported annual compensation is $65,000 among respondents with valid data.
- Implications: Findings support evidence-based technology training, recruitment and workforce planning.

INTRODUCTION



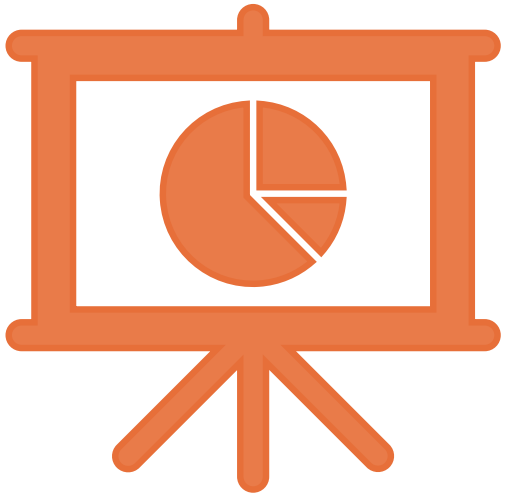
- Purpose: Analyse current technology adoption, future developer preferences and job-market signals.
- Target audience: Hiring teams, technology leaders, developers and learning & development specialists.
- Objectives: Compare current and desired languages/databases; examine jobs, compensation and demographics.
- Business value: Support recruitment, technical training and evidence-based workforce planning.

METHODOLOGY



- IBM capstone Jobs API dataset (JSON; 27,005 records)
- Python requests to retrieve job records
- Local filtering and counting by location and technology
- Excel exports using openpyxl
- Developer survey data prepared in Google Sheets
- Visualisations created in Google Looker Studio
- Web scraping: requests + BeautifulSoup parsed HTML table; exported 10 language/salary records to popular-languages.csv
- Data Limitations: Keyword-based matching may produce false positives, affecting the accuracy of technology demand estimates.

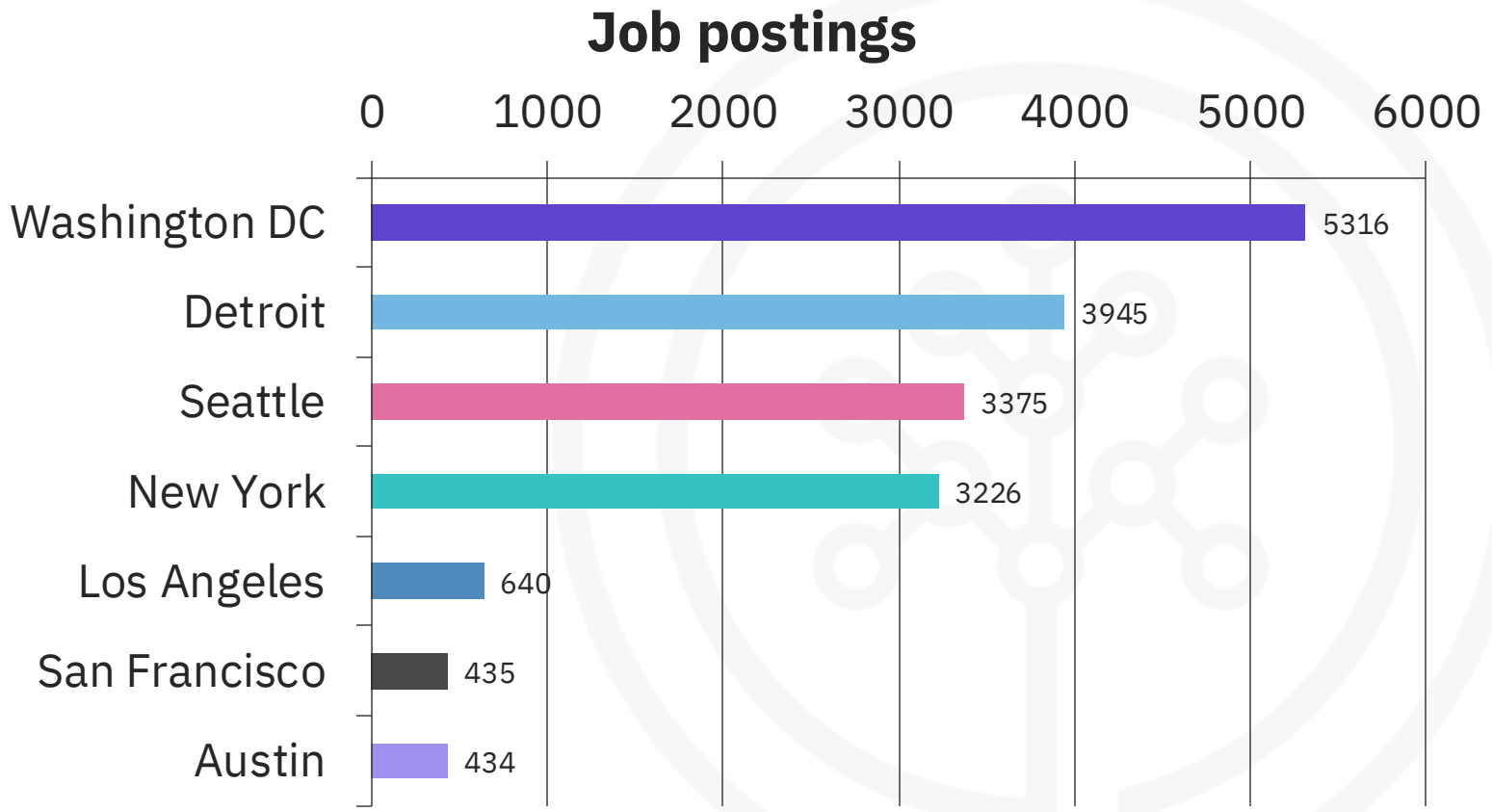
Results



Results presented in the following slides:

Job Postings • Popular Languages • Technology Trends

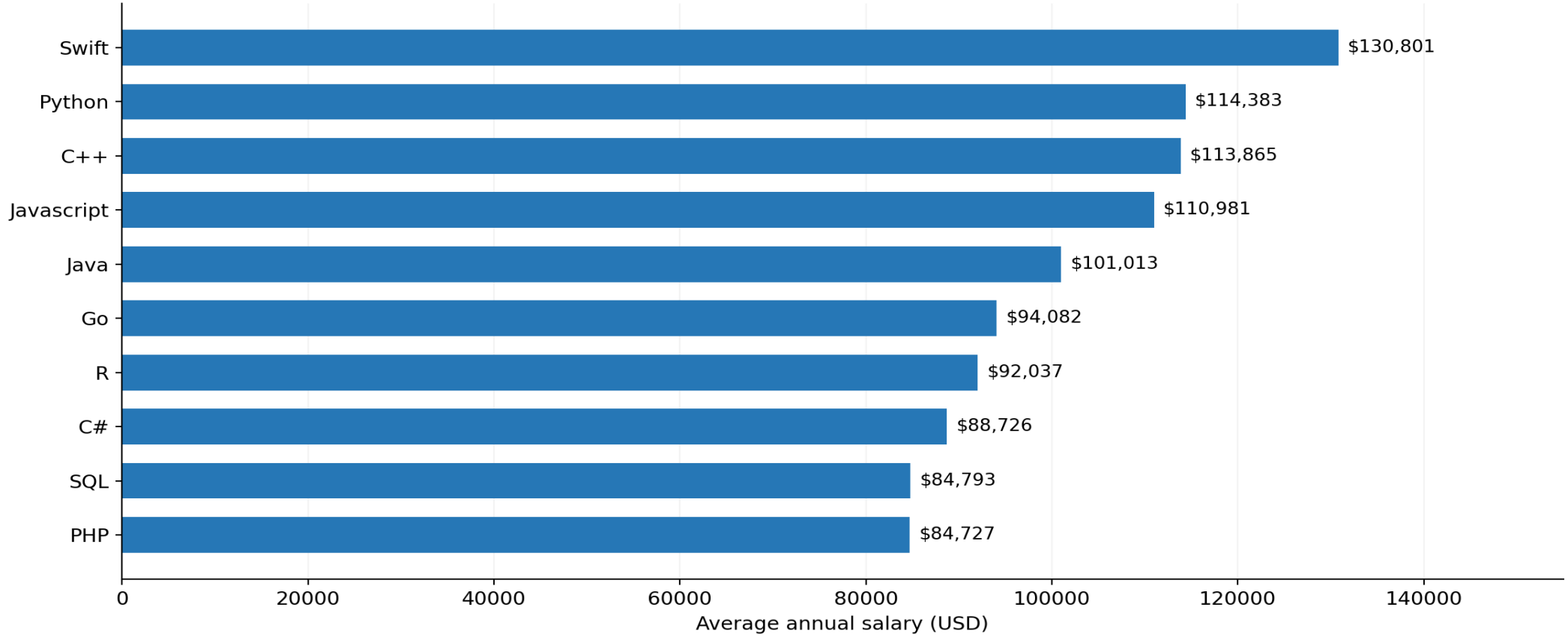
JOB POSTINGS



Source: IBM capstone lab Jobs API dataset (27,005 records); counts supplied from Module 1.

Highest: Washington DC (5,316) • Lowest: Austin (434)

POPULAR LANGUAGES

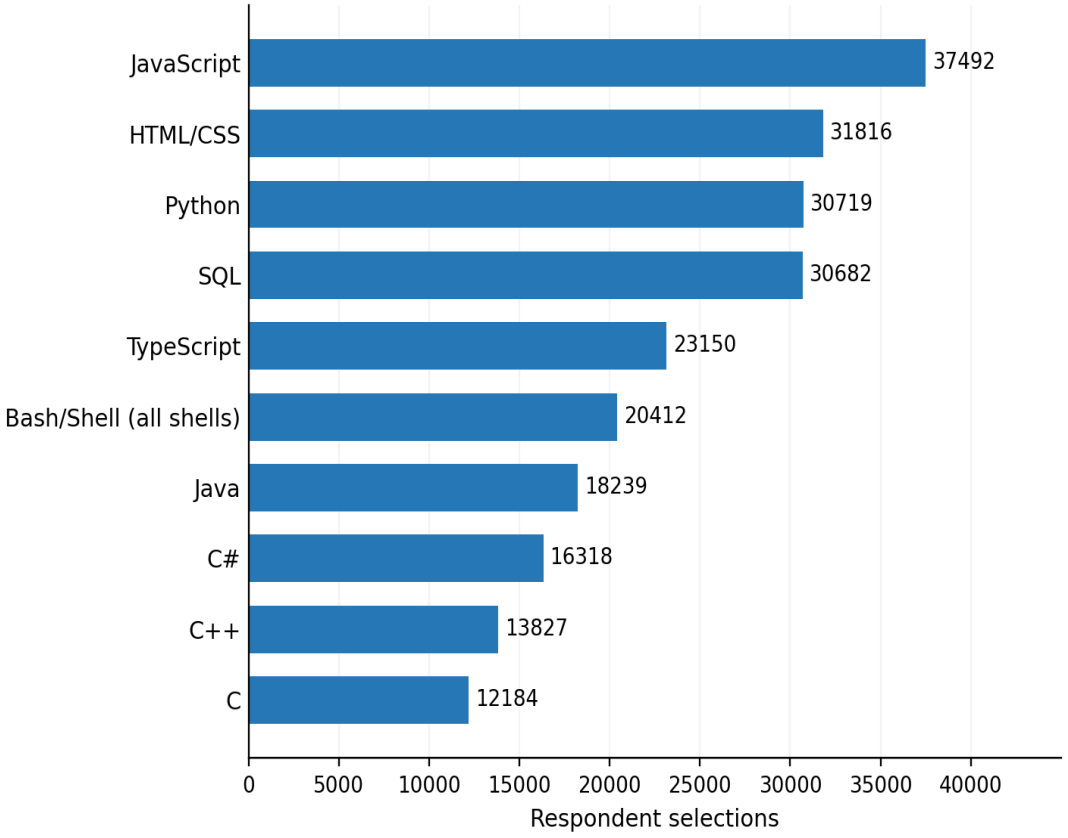


Source: Module 1 popular-languages.csv (historical training dataset). Salary values are not current market estimates.

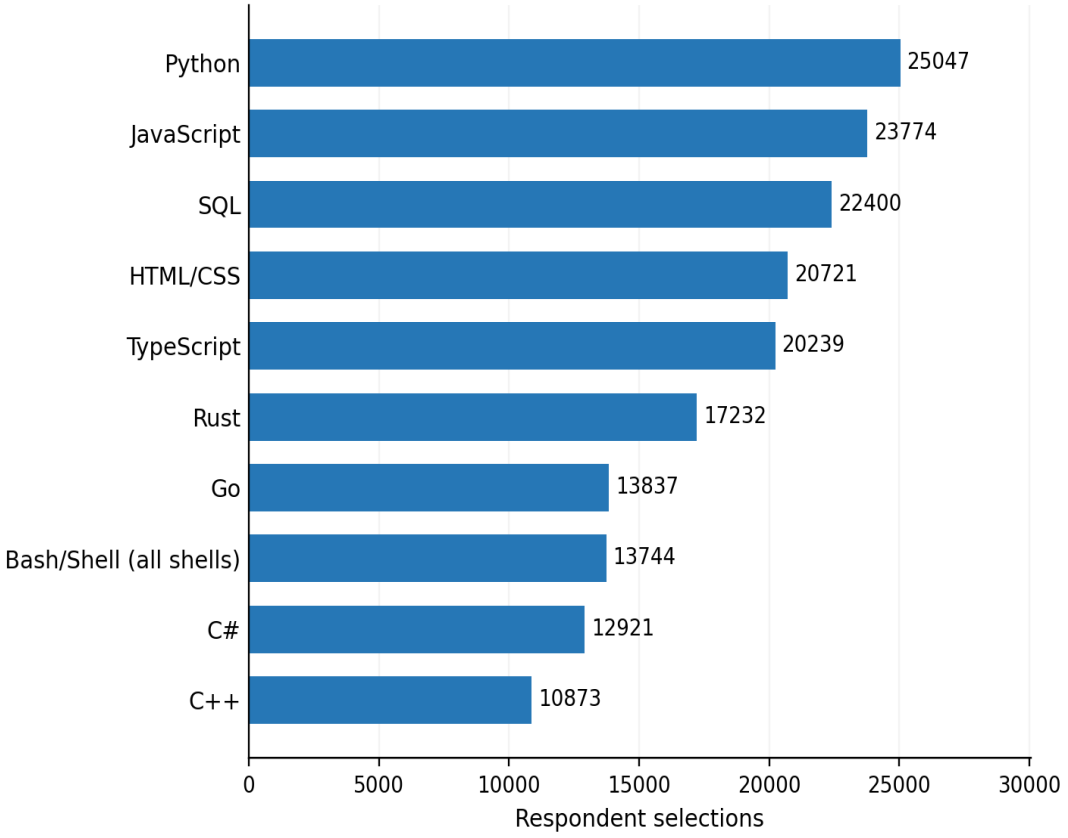
PROGRAMMING LANGUAGE TRENDS

Top 10 languages: currently used vs desired for future use (multi-select survey).

CURRENTLY USED



DESIRED FOR FUTURE USE

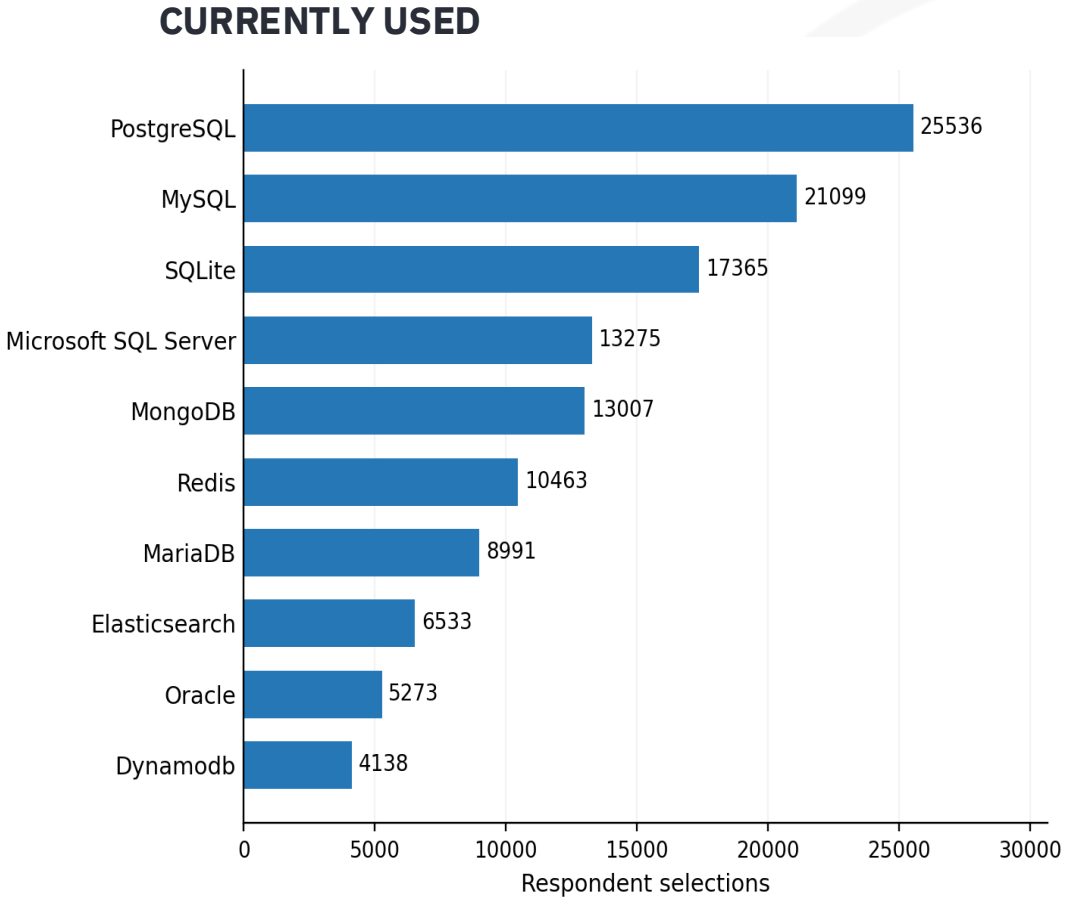


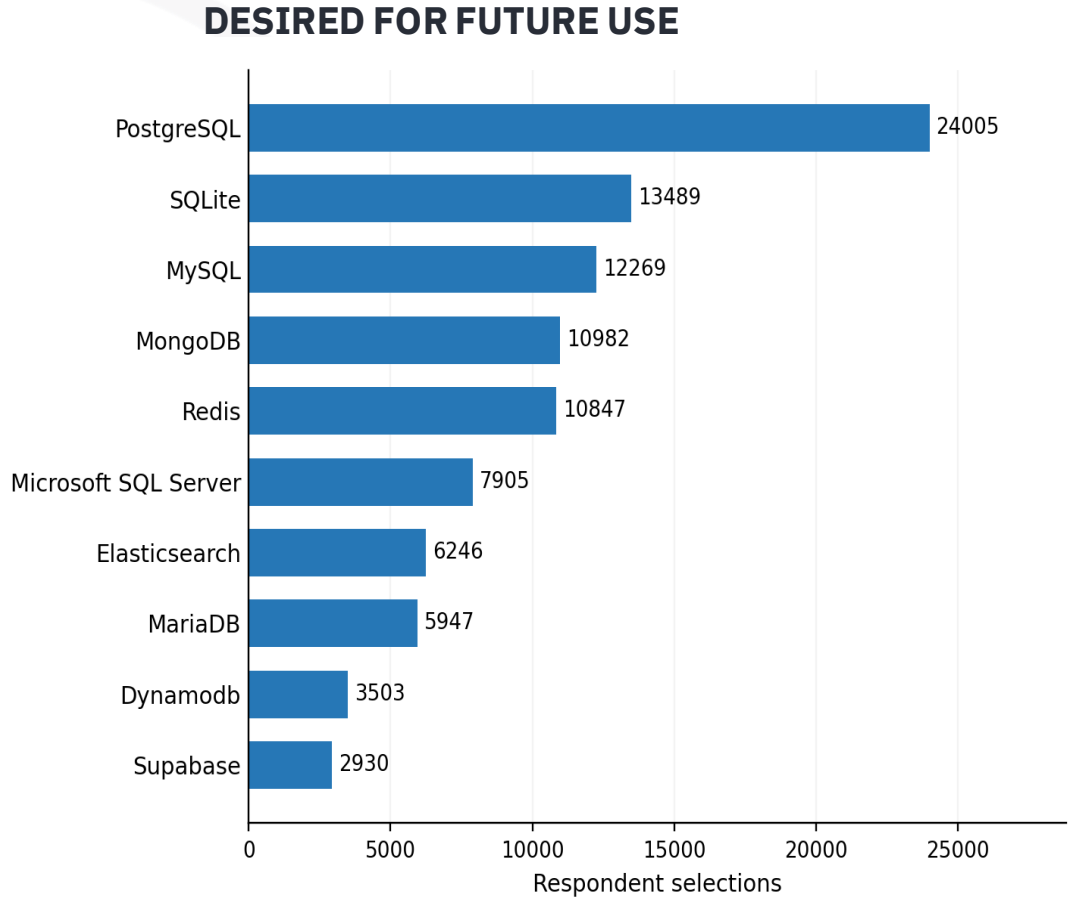
PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

- Findings: JavaScript leads current use (37,492 selections), while Python leads desired use (25,047). SQL appears in both top-ten rankings.
- Implications: Prioritise JavaScript and SQL for existing workflows, and Python for skills development. Desired selections express interest—not guaranteed future adoption.
- Prioritise JavaScript and SQL for widely used development workflows.

DATABASE TRENDS

Top 10 databases: currently used vs desired for future use (multi-select survey).





DATABASE TRENDS - FINDINGS & IMPLICATIONS

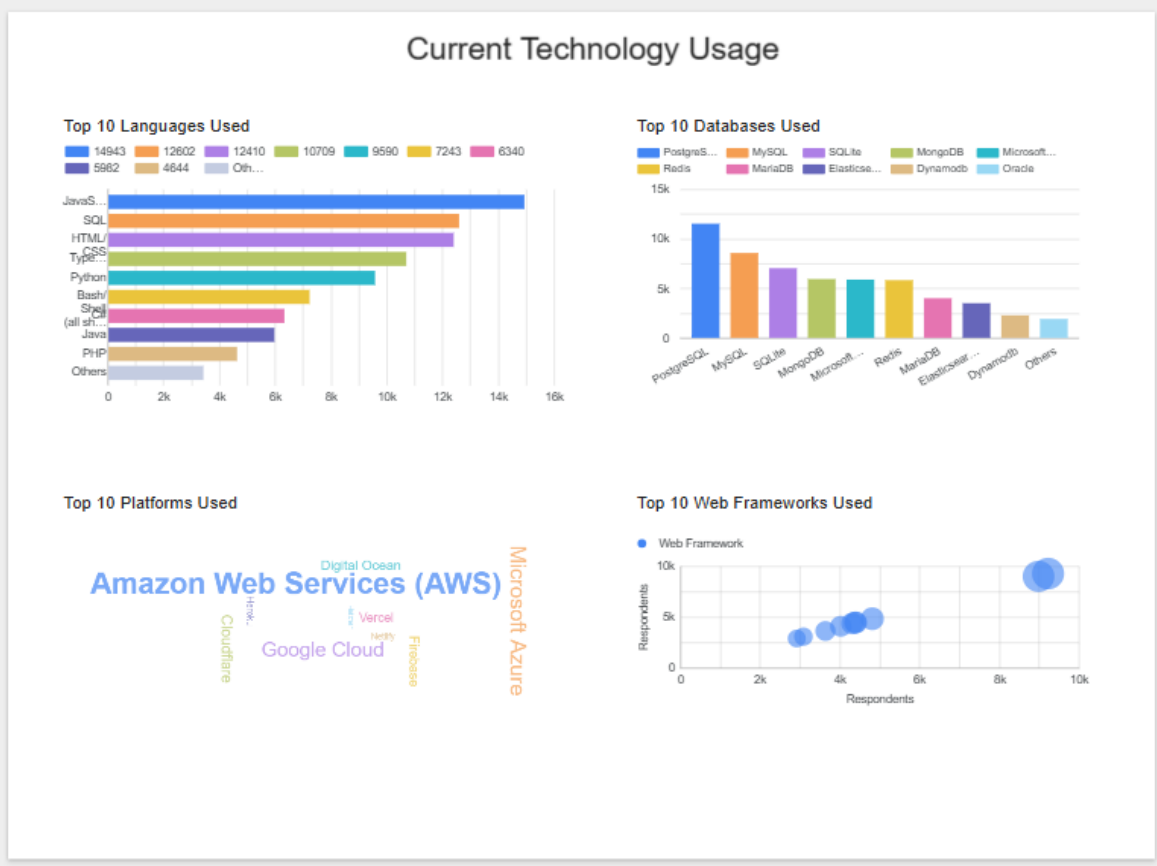
- Findings: PostgreSQL ranks first in current use (25,536) and desired use (24,005). MySQL and SQLite also appear among leading databases.
- Implications: Develop PostgreSQL and transferable SQL skills; compare survey preferences with organisational requirements before investing.
- PostgreSQL is a practical priority for database skills.

DASHBOARD



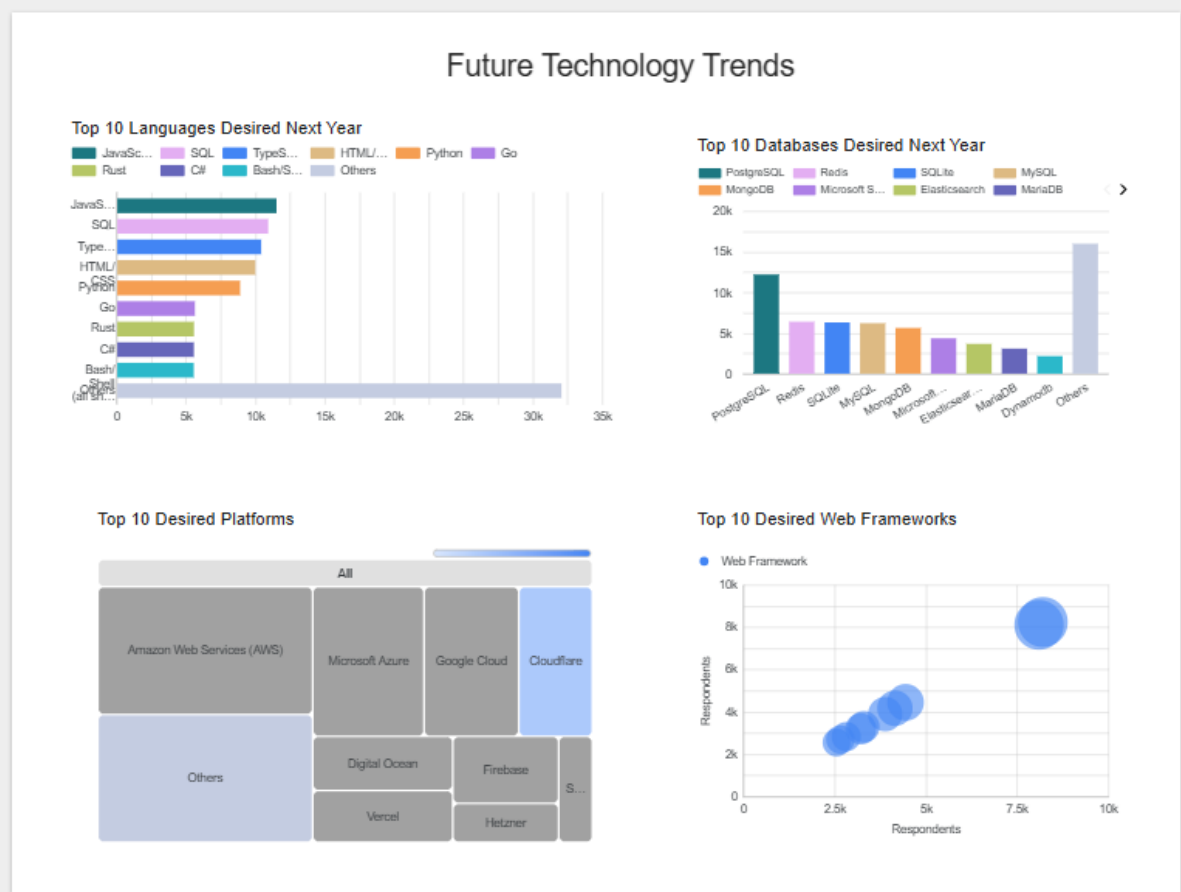
- Three interactive dashboards:
- Current Technology Usage
- Future Technology Trends
- Demographics

DASHBOARD TAB 1: Current Technology Usage



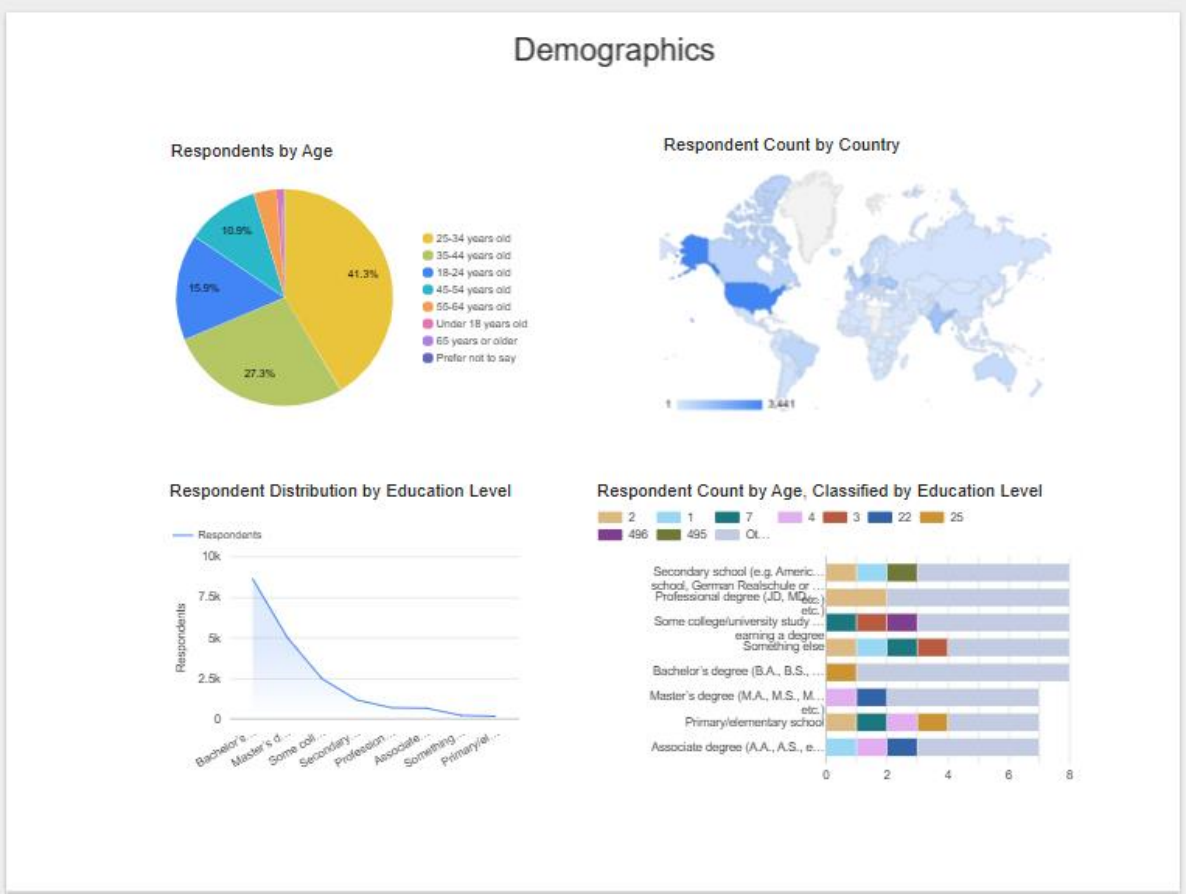
Key insight: JavaScript leads the language chart, followed by SQL and HTML/CSS. PostgreSQL leads database usage; AWS is prominent among platforms.

DASHBOARD TAB 2: Future Technology Trends



Key insight: JavaScript, SQL and TypeScript rank highly among named desired languages. PostgreSQL leads named desired databases; AWS, Azure and Google Cloud remain prominent.

DASHBOARD TAB 3: Demographics



Key insight: The largest age group is 25–34 (41.3%), followed by 35–44 (27.3%). Respondents are geographically distributed, with the US prominent on the map.

DISCUSSION



- The survey shows continuity in widely used technologies and shifts in expressed interest. JavaScript leads current language use, while Python leads desired languages. PostgreSQL leads both database measures. Dashboard demographics provide context for the respondent mix; selection bias and multiple-response questions limit generalisation.

OVERALL FINDINGS & IMPLICATIONS

- FINDINGS
- 1. JavaScript leads current language use; Python leads desired use.
- 2. PostgreSQL ranks first for both used and desired databases.
- 3. Survey respondents are concentrated in ages 25–44; technology preferences vary by population.

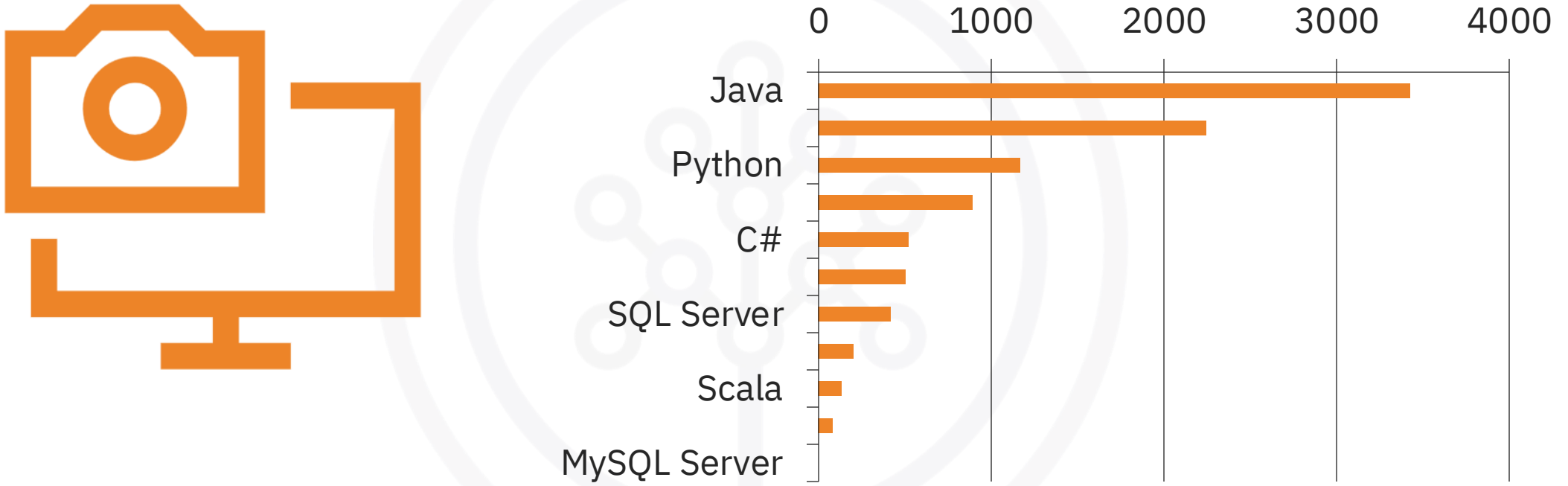
- INDUSTRY IMPLICATIONS
- 1. Maintain JavaScript/SQL capabilities while investing in Python upskilling.
- 2. Prioritise transferable SQL and PostgreSQL training for data-intensive roles.
- 3. Interpret survey demand in demographic context; validate with real job-market data.

CONCLUSION



- Technology Trends: JavaScript leads current use, while Python leads desired programming languages.
- Database Trends: PostgreSQL ranks first in both current and desired database selections.
- Workforce Insights: Developer demographics and technology preferences can inform recruitment and training.
- Business Value: Visualisations and dashboards support evidence-based technology and skills planning.

APPENDIX



Module 1 — Technology job counts (lab dataset). C omitted: substring matching returned 25,114 and overcounts unrelated skills. Other counts also use substring matching.

Examples of keyword matches: Java 3,428; JavaScript 2,248; Python 1,173. These are counts in the supplied historical lab data, not live vacancies.

APPENDIX — WEB SCRAPING LAB

- Objective: extract programming language and average annual salary from IBM-provided HTML table.
- Method: requests.get() → BeautifulSoup(html.parser) → table rows → CSV export.
- Output: popular-languages.csv, 10 languages; salary values are USD and reflect the source survey.
- Top 3 in source: Swift $130,801; Python $114,383; C++ $113,865.
- Source: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/datasets/Programming_Languages.html
- Caution: historical course sample, not verified current salary market data.

APPENDIX — EXPLORING THE DATASET

Purpose: profile the developer survey dataset before cleaning and analysing technology trends.

IBM lab workflow

1. Load survey-data-with-duplicate.csv with pandas.read_csv().
2. Inspect first five rows, row count, column count and column data types.
3. Map age brackets to approximate ages (17, 21, 29, 39, 49, 59, 70).
4. Calculate mean mapped age and number of distinct countries.

Data integrity note: the specific “with-duplicate” CSV was not accessible for independent execution. Numerical outcomes are deliberately not claimed; the accompanying notebook calculates them when the exact file is available. The mapped age is an approximation, not actual respondent age.

MODULE 3 — SURVEY CLEANING & EDA

65,457 raw rows → 65,437 unique responses | 20 duplicate records removed

Most-used languages
JavaScript: 37,492
HTML/CSS: 31,816
Python: 30,719
SQL: 30,682
TypeScript: 23,150

Most-desired languages
Python: 25,047
JavaScript: 23,774
SQL: 22,400
HTML/CSS: 20,721
TypeScript: 20,239

Compensation: median $65,000; IQR rule flags 978 extreme values among 23,435 reported salaries.
Interpret with care: many respondents did not report salary or job satisfaction.

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Source: user-provided survey_data_with_duplicate.csv | Unique ResponseId retained

MODULE 4 — VISUALISATION LABS

Labs 18–24 cover box, scatter, bubble, pie, stacked, line and bar charts.

Survey analysis used the deduplicated developer survey (65,437 responses). Technology counts split semicolon-separated multiple-choice selections and rank the ten most common values.

Interpretation: “Want to work with” measures expressed preference, not next-year adoption. Compensation and job-satisfaction charts are lab exercises and are not represented here as completed results.